

Procedure

Safety Note: To complete this lab you will need to collect water from a local pond or small lake. You should never enter a body of water alone. You may work with a partner for this lab and notify your instructor ahead of time of who you will be working with. If you don't work with a partner you should have a friend, parent or other individual accompany you on your visit to the pond to observe the pond and collect water. You should wear latex or rubber gloves when handling and collecting raw pond water.

1. Locate a suitable pond (or small lake) to investigate. Ask around if you are unsure. Many local parks have ponds that will be suitable for your use. Often, local individuals may know of areas in wooded ecosystems that are home to very nice pond for evaluating. If you aren't sure, talk to your mentor, a teacher at your school, or your parents to help locate a pond. Note that you should be able to collect some water from the pond safely and easily. Ponds that are surrounded by vegetation and growth may not be ideal. A pond at a park or elsewhere with a dock, swimming area, or boat launch would be ideal. Once you have identified a pond, record basic information about the pond in part 1 of the 9.1 Lab Student Response Sheet document.
2. Once you have identified a pond ecosystem to evaluate, make a plan to visit the pond. Once there, begin by making observations about the physical appearance of the pond and areas around the pond. Record your observations in the Physical Appearance Observations Table provided in part 2 of the 9.1 Lab Student Response Sheet.
3. Next, record biological characteristics of the pond by following the instructions below to search for both invertebrates and vertebrates within the ecosystem. Record all organisms found in the table in part 3 of the 9.1 Lab Student Response Sheet.
 - a) Use a D net to capture organisms in the littoral and benthic zones of the pond. Move the net along the bottom or the edge of the pond and empty the contents onto a white tray for easier identification. Repeat this process several times. Use a pond guide to key out the organisms you find. Be sure that you do not sample in the same spot and recapture what has already been inventoried. You may want to take pictures at the pond to study more closely later on. You may also want to use a microscope (if you have access to one) to search more closely for microorganisms.
 - b) Search the pond and shoreline for larger organisms or evidence of larger organisms. Reptiles, amphibians, birds, mammals, and fish could all be found. In addition to

looking for these organisms themselves, look for footprints, holes, bones etc that may indicate their presence. You can also look under logs, or rocks around the shoreline. If you do this be careful not to disrupt the ecosystem, be safe and return everything to its original location. Use field guides or internet searches to identify organisms. Again, you may want to photograph organisms to identify later after you return home.

- c) Finally, search the pond and shoreline for plants. Plants could be in the pond (e.g. lily pads, seaweed, algae) or along the shore (e.g. cat tails, grass, trees). Record the plants you observed, identifying each of them to the best of your ability.
4. Next, conduct at least 3 of the tests listed below to assess the chemical characteristics of the pond. You will need to follow the instructions provided with whatever water test kit you purchased to complete these tests. You may conduct them at the pond or you may want to bring a small sample of water back to your house or school using a bucket or jar for further study. Be sure that you have permission from the owner of pond or that the park allows this removal of a small amount of water. These two videos provide some examples for easy and simple ways to collect water samples for testing: [How to Take a Pond Water Sample](#) from the University of Florida and [Water Sampling](#) from Iowa State University. Remember to be safe around the pond, use appropriate equipment and have someone with you to watch and help. You will record the results of your tests in part 4 of the 9.1 Lab Student Response Sheet. You should complete AT LEAST 3 of the following tests:
- Coliform bacteria
 - Dissolved Oxygen
 - Nitrate
 - pH
 - Phosphate
 - Turbidity (Note that you can probably conduct a turbidity test without any specialized equipment or testing tools by following the description here:).